

Question 1

- Create 2 Public Docker Hub registries named
cloudethix_master_nginx_yourname & cloudethix_release_nginx_yourname.
- Clone below repository on your system.
Source: <https://github.com/zembutsu/docker-sample-nginx.git>
- Initialize a local repository & copy the code from above repo to your local repository in master branch and then create below branches.
release
main
hotfix
- Once code is copied to local repository, from master branch update the index.html and add word "Cloudethix Master Branch Nginx" and build the docker image & add meaningful tags and push to Docker Hub registry cloudethix_master_nginx_yourname.
- Also from release branch update the index.html and add word "Cloudethix Release Branch Nginx" and build the docker image & add meaningful tags and push to Docker Hub registry cloudethix_release_nginx_yourname.
- Once Images are copied to Docker hub registries, switch to the main branch.
- In main branch create directory named kube/clusterIP & inside kube directory create file named master_pod.yaml with pod name master_nginx & with label master_nginx & add image that you have pushed in Docker Hub registry cloudethix_master_nginx_yourname.
- Also create a file release_pod.yaml with pod name release_nginx & with label release_nginx & add image that you have pushed in Docker Hub registry cloudethix_release_nginx_yourname.
- Create a file called cluster_ip-service.yaml with service name cloudethix_clusterip and with Type clusterIP.
- Then, select the pod with label release_nginx in service.
- Create all these three resources in your k8s cluster.
- Now, access master_nginx pod shell & curl the master_nginx pod & check the result.
- Also try to curl release_nginx pod with DNS name & check the result.
- Then curl the clusterip service with its name and check the result.
- Finally, create a GITHUB remote repository named cloudethix-k8s-yourname and push all the branches to the remote repository.

```
achal@LAPTOP-# mkdir ~/cloudethix-k8s-achal
```

```
achal@LAPTOP-# cd ~/cloudethix-k8s-achal
```

```
achal@LAPTOP-# git init
```

```
hint: Using 'master' as the name for the initial branch. This default branch name
```

```
hint: will change to "main" in Git 3.0. To configure the initial branch name
```

```
hint: to use in all of your new repositories, which will suppress this warning,
```

```
hint: call:
```

```
hint:
```

```
hint:  git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:  git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in /home/achal/cloudethix-k8s-achal/.git/

-----

achal@LAPTOP-# git branch -M master

achal@LAPTOP-# cp -r ~/docker-sample-nginx/. .

achal@LAPTOP-HPUOTJF1:~/cloudethix-k8s-achal$ git branch

      hotfix
      main
* master
      release

achal@LAPTOP-HPUOTJF1:~/cloudethix-k8s-achal$ cat index.html
```

Cloudethix Master Branch Nginx

Hostname: nginx

```
achal@LAPTOP-#:~/cloudethix-k8s-achal$ podman build -t cloudethix-master-nginx:master-v1 .

STEP 1/3: FROM nginx:alpine
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
--> Using cache
090b497e1161956bc69c29ed05f63003fabe8615f06a8b8a09cc43f4e2783826
--> 090b497e1161
STEP 3/3: COPY index.html /usr/share/nginx/html/
COMMIT cloudethix-master-nginx:master-v1
--> 0a87466d9604
Successfully tagged localhost/cloudethix-master-nginx:master-v1
0a87466d96047a9e8a7f0d0bef83e0accd0c1116e76f7cc8e68342953b56badc
```

achal@LAPTOP-#:~/cloudethix-k8s-achal\$ podman images			
REPOSITORY		TAG	IMAGE ID
CREATED	SIZE		
localhost/cloudethix-master-nginx		master-v1	0a87466d9604
17 seconds ago	64.2 MB		

```
achal@LAPTOP-#:~/cloudethix-k8s-achal$ podman tag cloudethix-master-nginx:master-v1 \
docker.io/achalpadol/cloudethix_master_nginx_achal:master-v1
```

```
-----
achal@LAPTOP-#:~/cloudethix-k8s-achal$ podman push
docker.io/achalpadol/cloudethix_master_nginx_achal:master-v1
Getting image source signatures
Copying blob 0d62d88506ba skipped: already exists
Copying blob 55afa1ecc21d skipped: already exists
Copying blob 1ed8e39a7434 skipped: already exists
Copying blob 8cdfef4f23778 skipped: already exists
Copying blob d94291c26261 skipped: already exists
Copying blob 0ea935727878 skipped: already exists
Copying blob 3b969d6574e5 done    |
Copying blob 0cc7f3a48f3f skipped: already exists
Copying blob da4dea1b00af skipped: already exists
Copying blob fb597529c916 skipped: already exists
Copying config 0a87466d96 done    |
Writing manifest to image destination
-----
```

```
achal@LAPTOP-# git switch release
Switched to branch 'release'
achal@LAPTOP-HPU0TJF1:~/cloudethix-k8s-achal$ git branch
```

```
    hotfix
    main
    master
* release
```

```
achal@LAPTOP-#:~/cloudethix-k8s-achal$ vim index.html
achal@LAPTOP-#:~/cloudethix-k8s-achal$ cat index.html
<!DOCTYPE html>
<html>
<head>
    <title>CloudEthiX Nginx</title>
</head>
<body>
    <h1>Cloudethix Release Branch Nginx</h1>
    <h2>Hostname: nginx</h2>

</body>
</html>
-----
```

```
achal@LAPTOP-# podman build -t cloudethix-release-nginx:release-v1 .
STEP 1/3: FROM nginx:alpine
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
--> Using cache
090b497e1161956bc69c29ed05f63003fabe8615f06a8b8a09cc43f4e2783826
--> 090b497e1161
STEP 3/3: COPY index.html /usr/share/nginx/html/
COMMIT cloudethix-release-nginx:release-v1
--> 27988c647986
Successfully tagged localhost/cloudethix-release-nginx:release-v1
27988c6479868568ab43601469aa0a8c29727415229be9e09d751c5f278b82f4
-----
```

```
achal@LAPTOP-# podman tag cloudethix-release-nginx:release-v1 \
docker.io/achalpadol/cloudethix_release_nginx_achal:release-v1
-----
```

```
achal@LAPTOP-#:~/cloudethix-k8s-achal$ podman push
docker.io/achalpadol/cloudethix_release_nginx_achal:release-v1
Getting image source signatures
Copying blob 1ed8e39a7434 skipped: already exists
Copying blob 0ea935727878 skipped: already exists
```

```
Copying blob 55afa1ecc21d skipped: already exists
Copying blob d94291c26261 skipped: already exists
Copying blob 8cdfe4f23778 skipped: already exists
Copying blob 0d62d88506ba skipped: already exists
Copying blob 8d7e9d98513f done    |
Copying blob fb597529c916 skipped: already exists
Copying blob da4dea1b00af skipped: already exists
Copying blob 0cc7f3a48f3f skipped: already exists
Copying config 27988c6479 done    |
Writing manifest to image destination
-----

achal@LAPTOP-#:~/cloudethix-k8s-achal$ git switch main
Switched to branch 'main'
achal@LAPTOP-HPU0TJF1:~/cloudethix-k8s-achal$ git branch
hotfix
* main
  master
  release
-----

achal@LAPTOP-# ls
cluster_ip-service.yaml  master_pod.yaml  release_pod.yaml
achal@LAPTOP-# cat cluster_ip-service.yaml release_pod.yaml master_pod.yaml

apiVersion: v1
kind: Service
metadata:
  name: cloudethix-clusterip
  namespace: cloudethix-achal

spec:
  type: ClusterIP

  selector:
    app: release-nginx

  ports:
    - protocol: TCP
      port: 80
      targetPort: 80

-----

apiVersion: v1
kind: Pod
metadata:
  name: release-nginx
  namespace: cloudethix-achal
  labels:
    app: release-nginx

spec:
  containers:
    - name: release-nginx
      image: docker.io/achalpadol/cloudethix_release_nginx_achal:release-v1
      ports:
        - containerPort: 80

-----

apiVersion: v1
```

```
kind: Pod
metadata:
  name: master-nginx
  namespace: cloudethix-achal
  labels:
    app: master-nginx

spec:
  containers:
    - name: master-nginx
      image: docker.io/achalpadol/cloudethix_master_nginx_achal:master-v1
      ports:
        - containerPort: 80
  -----
```

```
achal@LAPTOP-#:~/cloudethix-k8s-achal/kube/clusterIP$ kubectl get pods -n cloudethix-achal
```

NAME	READY	STATUS	RESTARTS	AG
cloudethix-nginx-5d77d6699f-ktvsq	1/1	Running	0	87m
master-nginx	1/1	Running	0	21s
release-nginx	1/1	Running	0	13s

```
chal@LAPTOP-# kubectl exec -it master-nginx -n cloudethix-achal -- /bin/sh
/ # curl http://localhost:80
```

```
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX Nginx</title>
</head>
<body>
  <h1>Cloudethix Master Branch Nginx</h1>
  <h2>Hostname: nginx</h2>

</body>
</html>
```

```
/ # hostname
```

```
master-nginx
```

```
achal@LAPTOP-# kubectl get pod release-nginx -n cloudethix-achal -o wide
NAME          READY  STATUS    RESTARTS  AGE  IP          NODE
NOMINATED NODE  READINESS GATES
release-nginx  1/1    Running   0          17m  10.244.0.97 minikube
<none>        <none>
```

```
achal@LAPTOP-#1:~/cloudethix-k8s-achal/kube/clusterIP$ kubectl exec -it master-nginx -n cloudethix-achal -- /bin/sh
```

```
/ # curl http://cloudethix-clusterip
```

```
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX Nginx</title>
</head>
<body>
  <h1>Cloudethix Release Branch Nginx</h1>
```

```
<h2>Hostname: nginx</h2>
```

```
</body>
```

```
</html>
```

Question 2

- In the main branch of your local repository create a directory

kube/NodePort.

- Create below files from below url. Please make sure you will create NodePort service with port 30008 instead of loadbalancer.

Source: <https://kubernetes.io/docs/tasks/access-application-cluster/connec>

ting-frontend-backend/.

backend-deployment.yaml

backend-service.yaml

frontend-deployment.yaml

frontend-NodePort-service.yaml

- Once files are created , create all the resources in your k8s cluster.

- Access all public ips with port 30008 in the browser and then check the result.

- Finally, push all the latest code to the remote repository.

- Take all screenshots and create a well formatted document

```
```text
```

```
achal@LAPTOP-# cat backend-deployment.yaml backend-service.yaml frontend-NodePort-service.yaml frontend-deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
 name: backend
```

```
 namespace: cloudethix-achal
```

```
spec:
```

```
 replicas: 3
```

```
 selector:
```

```
 matchLabels:
```

```
 app: hello
```

```
 tier: backend
```

```
 track: stable
```

```
 template:
```

```
 metadata:
```

```
 labels:
```

```
 app: hello
```

```
 tier: backend
```

```
 track: stable
```

```
 spec:
```

```
 containers:
```

```
 - name: hello
```

```
 image: gcr.io/google-samples/hello-go-gke:1.0
```

```
 ports:
 - name: http
 containerPort: 80
```

---

```
apiVersion: v1
kind: Service
metadata:
 name: hello
 namespace: cloudethix-achal
```

```
spec:
 selector:
 app: hello
 tier: backend

 ports:
 - protocol: TCP
 port: 80
 targetPort: http
```

---

```
apiVersion: v1
kind: Service
metadata:
 name: frontend
 namespace: cloudethix-achal
```

```
spec:
 type: NodePort

 selector:
 app: hello
 tier: frontend

 ports:
 - protocol: TCP
 port: 80
 targetPort: 80
 nodePort: 30008
```

---

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: frontend
 namespace: cloudethix-achal
```

```
spec:
 replicas: 1

 selector:
 matchLabels:
 app: hello
 tier: frontend
 track: stable
```

```
template:
 metadata:
 labels:
 app: hello
```

```

 tier: frontend
 track: stable

spec:
 containers:
 - name: nginx
 image: gcr.io/google-samples/hello-frontend:1.0

 lifecycle:
 preStop:
 exec:
 command:

 - /usr/sbin/nginx
 - -s
 - quit

```

```

achal@LAPTOP-# kubectl get pod -n cloudehix-achal
NAME READY STATUS RESTARTS AGE
backend-7cdc9fcc87-ghbgx 1/1 Running 0 54s
backend-7cdc9fcc87-mrhzf 1/1 Running 0 54s
backend-7cdc9fcc87-n9sfh 1/1 Running 0 54s
frontend-7db75b888d-w6jdn 1/1 Running 0 24s

```

```

achal@LAPTOP-# kubectl get svc frontend -n cloudehix-achal
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
frontend NodePort 10.106.74.6 <none> 80:30008/TCP 25m

```

```

achal@LAPTOP-# $ minikube service frontend -n cloudehix-achal --url

http://127.0.0.1:33451
! Because you are using a Docker driver on linux, the terminal needs to be
open to run it.

```

=====  
Question 3

```

- Create any 2 pods and assign them to different worker nodes with
nodeName property.

```

```

```text
achal@LAPTOP-#$ cat pod-node02.yaml pod-node04.yaml

apiVersion: v1
kind: Pod
metadata:
  name: pod-node-02
  namespace: cloudehix-achal
spec:
  nodeName: minikube-m02
  containers:
    - name: nginx
      image: nginx:latest
      ports:

```



```
- containerPort: 80
-----
apiVersion: v1
kind: Pod
metadata:
  name: pod-node04
  namespace: cloudethix-achal
spec:
  nodeName: minikube-m04
  containers:
    - name: nginx
      image: nginx:latest
      ports:
        - containerPort: 80
-----
achal@LAPTOP-# kubectl get pods -o wide -n cloudethix-achal
NAME                                READY   STATUS    RESTARTS   AGE   IP
NODE                                READINESS GATES
pod-node-02                         1/1     Running   0           101s
10.244.1.139   minikube-m02   <none>     <none>
pod-node04                         1/1     Running   0           31s
10.244.2.2     minikube-m04   <none>     <none>
```

=====

Question 4

- Label both worker nodes such as worker-0 node as cloudethix-k8s-00 & worker-1 node as cloudethix-k8s-01.
- Once nodes are labeled, create pod00.yaml file and schedule the pod on worker-0 node with nodeSelector property. Also create one more file named pod01.yaml & schedule the pod on worker-1 node.

=====

```
```text
achal@LAPTOP-# kubectl label node minikube-m02 cloudethix-k8s-00=worker-0
node/minikube-m02 labeled

achal@LAPTOP-# kubectl label node minikube-m04 cloudethix-k8s-01=worker-1
node/minikube-m04 labeled

achal@LAPTOP-# kubectl get nodes --show-labels
NAME STATUS ROLES AGE VERSION LABELS
minikube Ready control-plane 14d v1.35.1
beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube,kubernetes.io/os=linux,minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=true,minikube.k8s.io/updated_at=2026_08_18T12_18_49_0700,minikube.k8s.io/version=v1.38.1,node-role.kubernetes.io/control-plane=,node.kubernetes.io/exclude-from-external-load-balancers
minikube-m02 Ready <none> 14d v1.35.1
beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,cloudethix-k8s-00=worker-0,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube-m02,kubernetes.io/os=linux,minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=false,minikube.k8s.io/updated_at=2026_08_18T12_18_58_0700,minikube.k8s.io/version=v1.38.1
minikube-m04 Ready <none> 63m v1.35.1
beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,cloudethix-k8s-01=worker-1,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube-
```

```
m04,kubernetes.io/os=linux,minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=false,minikube.k8s.io/updated_at=2026_09_02T07_45_47_0700,minikube.k8s.io/version=v1.38.1
```

```

achal@LAPTOP-# cat pod00.yaml pod01.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
 name: pod00
 namespace: cloudethix-achal
spec:
 nodeSelector:
 cloudethix-k8s-00: worker-0
 containers:
 - name: nginx
 image: nginx:latest
 ports:
 - containerPort: 80
```

```

apiVersion: v1
kind: Pod
metadata:
 name: pod01
 namespace: cloudethix-achal
spec:
 nodeSelector:
 cloudethix-k8s-01: worker-1
 containers:
 - name: nginx
 image: nginx:latest
 ports:
 - containerPort: 80
```

```

achal@LAPTOP-# kubectl get pods -o wide -n cloudethix-achal
```

NAME		READY	STATUS	RESTARTS	AGE	IP
NODE	NOMINATED NODE	READINESS GATES				
pod00		1/1	Running	0	2m20s	
10.244.1.140	minikube-m02	<none>	<none>			
pod01		1/1	Running	0	50s	
10.244.2.3	minikube-m04	<none>	<none>			

=====  
  
Question 5

- Clone the below repo locally & create DaemonSet from directory DaemonSet101.  
Source: <https://github.com/collabnix/kubelabs>

```
=====

```text  
achal@LAPTOP-HPUOTJF1:~$ cd ~  
git clone https://github.com/collabnix/kubelabs.git  
Cloning into 'kubelabs'...  
remote: Enumerating objects: 14031, done.
```

remote: Counting objects: 100% (707/707), done.
remote: Compressing objects: 100% (147/147), done.
remote: Total 14031 (delta 637), reused 560 (delta 560), pack-reused 13324 (from 5)
Receiving objects: 100% (14031/14031), 61.81 MiB | 9.10 MiB/s, done.
Resolving deltas: 100% (4181/4181), done.

achal@LAPTOP-HPUOTJF1:~\$ cd ~/kubelabs/DaemonSet101
achal@LAPTOP-HPUOTJF1:~/kubelabs/DaemonSet101\$ ls -la
total 12

-rw-r--r-- 1 achal achal 7040 Sep 2 09:01 README.md
-rw-r--r-- 1 achal achal 394 Sep 2 09:01 daemonset.yml

achal@LAPTOP-HPUOTJF1:~/kubelabs/DaemonSet101\$ cat daemonset.yml

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: prometheus-daemonset
  namespace: cloudethix-achal
spec:
  selector:
    matchLabels:
      tier: monitoring
      name: prometheus-exporter
  template:
    metadata:
      labels:
        tier: monitoring
        name: prometheus-exporter
    spec:
      containers:
        - name: prometheus
          image: prom/node-exporter
          ports:
            - containerPort: 80
```

chal@LAPTOP-HPUOTJF1:~/kubelabs/DaemonSet101\$ kubectl get daemonset -n cloudethix-achal

NAME	DESIRED	CURRENT	READY	UP-TO-DATE	AVAILABLE
prometheus-daemonset	3	3	3	3	3
<none>					

19s

=====

Question 6

- Create a static pod with name cloudethix-static in your k8s cluster. Refer below link.
Source: <https://kubernetes.io/docs/tasks/configure-pod-container/static-pod>

=====

```
```text
achal@LAPTOP-HPUOTJF1:~$ cat cloudethix-static-pod
apiVersion: v1
kind: Pod
metadata:
 name: cloudethix-static
 namespace: cloudethix-achal
 labels:
 role: myrole
spec:
 containers:
 - name: web
 image: nginx
 ports:
 - name: web
 containerPort: 80
 protocol: TCP
```

---

```
achal@LAPTOP-HPUOTJF1:~$ minikube cp cloudethix-static-pod
/etc/kubernetes/manifests/cloudethix-static.yaml
```

```
achal@LAPTOP-HPUOTJF1:~$ minikube ssh
Linux minikube 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon
Dec 1 20:46:23 UTC 2025 x86_64
```

The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/\*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.

---

```
docker@minikube:~$ cd /etc/kubernetes/manifests/
```

```
docker@minikube:/etc/kubernetes/manifests$ ls
cloudethix-static.yaml etcd.yaml kube-apiserver.yaml kube-controller-
manager.yaml kube-scheduler.yaml
```

---

```
achal@LAPTOP-HPUOTJF1:~$ kubectl get pod
```

NAME	READY	STATUS	RESTARTS	AGE
cloudethix-static-minikube	1/1	Running	0	43s

---

## Question 7

- Install Kubectx & kubens in your k8s cluster

---

```
```text
sudo apt update
sudo apt install kubectx -y
```

```
achal@LAPTOP-HPUOTJF1:~$ sudo apt install kubectx -y
kubectx is already the newest version (0.9.5-2build1).
Summary:
```

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 40

kubectx

kubens

kubectx minikube

kubens cloudethix-achal

kubectl get pods